

3(a)	(force =) rate of change of momentum	B1
3(b)(i)	$E = \frac{1}{2}mv^2$ or $\frac{1}{2} \times 0.062 \times 3.8^2$ or $\frac{1}{2} \times 0.062 \times 1.7^2$	C1
	loss of KE = $\frac{1}{2} \times 0.062 \times (3.8^2 - 1.7^2)$ = 0.36 J	A1
3(b)(ii)	$p = mv$ or 0.062×3.8 or 0.062×1.7	C1
	change in momentum = $0.062 \times (1.7 + 3.8)$ = 0.34 N s	A1
3(b)(iii)	(average resultant force =) $0.34 / 0.081 = 4.2$ (N) or (average resultant force =) $0.062 \times (1.7 + 3.8) / 0.081 = 4.2$ (N)	A1
3(b)(iv)	1. average force = $4.2 + (0.062 \times 9.81)$ = 4.8 N	A1
	2. average force = 4.8 N	A1